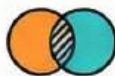


The Probability Decision Matrix: Mastering Board Exam 2025

Essential Foundation Formulas



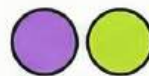
Bag A



Conditional Probability $P(A|B)$
 $P(A|B) = P(A \cap B) / P(B)$, where $P(B) \neq 0$



The Sum of Probabilities
For a valid probability distribution, the sum of all individual probabilities $\sum P(x)$ must always equal 1.



Mutual Exclusivity (Disjoint Events)
If events A and B are mutually exclusive, their intersection $P(A \cap B) = 0$, simplifying the Union formula to $P(A \cup B) = P(A) + P(B)$.



Bag B

Two-Bag Decision Matrix

Scenario: Bag A (6 Red, 4 Blue) | Bag B (5 Red, 2 Blue)

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Conditional Probability: "The Direct Path"

Select Bag A, then find the probability of a Red ball.



Theorem of Total Probability: "The Outcome Focus"

What is the probability of picking a Red ball?



Σ

You must sum all possible paths
(Bag A $\rightarrow$ Red + Bag B $\rightarrow$ Red)



Bayes' Theorem: "The Reverse Path"

A Red ball was picked; what is the probability it came from Bag A?



The outcome is known; you are seeking the cause/source.

Independent vs. Dependent Events

Independent Events (With Replacement)



$$P(E \cap F) = P(E) \cdot P(F)$$

When a ball is replaced after picking, the probability of the second pick remains unchanged.

Dependent Events (Without Replacement)

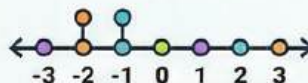


When a ball is NOT replaced, the first event changes the total count, affecting the second event's probability (Multiplication Theorem).

Random Variables & Distribution

Random Variable (X)

A variable that contains all points of a sample space; it can be positive, negative, or zero.



Mean / Expected Value $E(X)$

Calculated by multiplying each random variable value by its probability and summing them:

$$\mu = \sum x_i p_i$$



Variance and Standard Deviation

Measures of dispersion $(V(X) = E(X^2) - [E(X)]^2)$ that describe how much the probability distribution spreads from the mean.



A Visual Guide to Mastering Linear Programming Problems (LPP)

@ThunderStudy

1. Understand the Core Components

Z The Objective Function (Z)

$$Z = ax + by$$

↑
MAXIMIZE

↓
MINIMIZE

Decision Variables (x, y)



Variables you control
(Non-negative constraints: $x \geq 0, y \geq 0$)

2. Master Graphing the Constraints

Step 1: Draw the Line

$$3x + 6y \geq 12$$

$$3x + 6y = 12$$

Step 2: Choose Line Type: Solid vs. Dotted

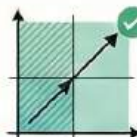
$\leq$ or $\geq$

Solid Line

$<$ or $>$

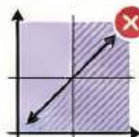
Dotted Line

Step 3: Determine Shading with the Origin Test (0, 0)



TRUE
(e.g., $0 \leq 12$)

→ Shade towards origin



FALSE
(e.g., $0 < 12$)

→ Shade away from origin

Special Case:
Line Passes Through Origin
(e.g., $x \geq y$)

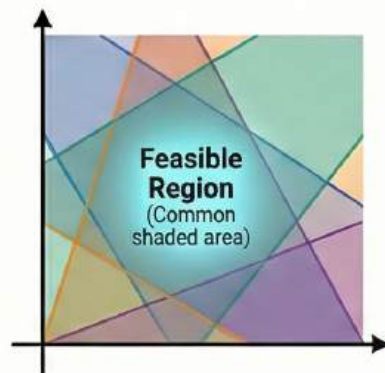


$x > y$

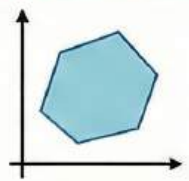


$y > x$

3. Identify the Feasible Region

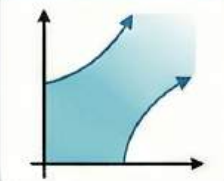


Bounded Region



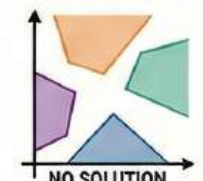
Completely enclosed
(Max & Min exist at corner points)

Unbounded Region



Open on at least one side
(Max/Min may or may not exist - Check required)

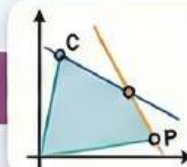
Infeasible Region



NO SOLUTION
No common shaded area

4. Find the Optimal Solution (Max/Min Value)

For Bounded Regions

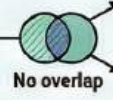
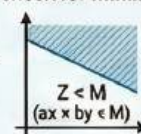


Test Z at each corner point.
Largest = Max,
Smallest = Min

For Unbounded Regions - A Crucial Check

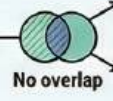
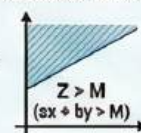
After finding potential Max/Min M:

To check for Minimum M:



NO MINIMUM VALUE
M is the MINIMUM VALUE

To check for Maximum M:

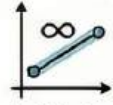


NO MAXIMUM VALUE
M is the MAXIMUM VALUE

5. Key Exam Facts to Remember



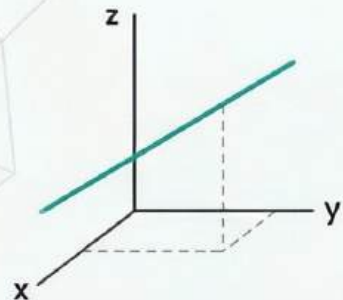
Solutions Can Occur at Multiple Points
(Max/Min can exist at more than one corner point)



Infinite Solutions on a Line Segment
(If two adjacent corner points give same optimal value, every point on the connecting segment is also an optimal solution)

3D Geometry Rapid Revision: Mastering the "Straight Line" for Class 12 Boards

Vector Algebra & 3D Geometry: The High-Value Pair (approx. 7-8 marks for 3D lines) **@ThunderStudy**



FUNDAMENTAL EQUATIONS OF A LINE

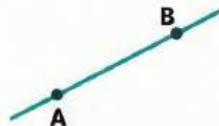
One Point Form



Vector Form:
as parallel vector:
 $\mathbf{r} = \mathbf{a} + \lambda \mathbf{m}$

Cartesian Form:
$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Two Point Form



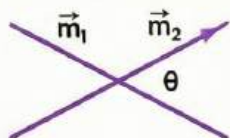
Vector Form:
as parallel vector:
 $(\mathbf{b} - \mathbf{a})$

Cartesian Form:
$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

THE STANDARD FORM WARNING:

Always ensure coefficients of x, y, z are ± 1 before extracting direction ratios.

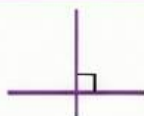
ANGLES AND LINE RELATIONSHIPS



Finding the Angle Between Two Lines

The angle is the same as between their parallel vectors using the dot product formula.

Condition for Perpendicularity



$$\mathbf{m}_1 \cdot \mathbf{m}_2 = 0$$

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Condition for Parallelism



Their direction ratios are proportional:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

SHORTEST DISTANCE (SD)



Skew Lines

Non-parallel, non-intersecting in 3D



Parallel Lines

Never intersect, coplanar

Distinguishing Skew vs. Parallel Lines:

Check if parallel vectors $\mathbf{m}_1$ & $\mathbf{m}_2$ are scalar multiples; if yes, parallel; if no, skew.

Formulas for Shortest Distance (d)

Skew Lines

$$d = \frac{|\mathbf{m}_1 \times \mathbf{m}_2 \cdot (\mathbf{a}_2 - \mathbf{a}_1)|}{|\mathbf{m}_1 \times \mathbf{m}_2|}$$

Parallel Lines

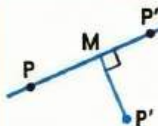
$$d = \frac{|\mathbf{m} \times (\mathbf{a}_2 - \mathbf{a}_1)|}{|\mathbf{m}|}$$

Intersecting Lines: If Shortest Distance is 0, lines are confirmed to be intersecting.

ADVANCED 'EXAM-SECRET' TOPICS



Finding the Point of Intersection
Equate general points (A and B), solve equations, and verify with the third coordinate.



Foot of Perpendicular and Image

Find general point M on the line, ensure direction PM is perpendicular (dot product = 0), then use midpoint formula to find the image.

DC set



Direction Cosines (DC) Properties

A line has two sets of DCs (+ and -); unique, unlike Direction Ratios which can be any proportional values.



Vector Algebra Rapid Revision: 2025 Board Exam Cheat Sheet

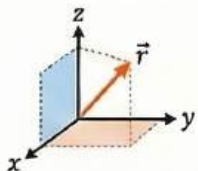
Fundamentals & Vector Types

What is a Vector?

A quantity that has both **magnitude** (length) and **direction**, represented as $\vec{a}$ or $\overrightarrow{AB}$ (from initial point A to terminal point B).

Magnitude Formula

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$



Essential Vector Types

 **Zero Vectors**
(initial/terminal points same)

$\hat{a}$ **Unit Vectors**
(magnitude = 1, denoted as $\hat{a}$)

 **Co-initial Vectors**
(starting from the same point)

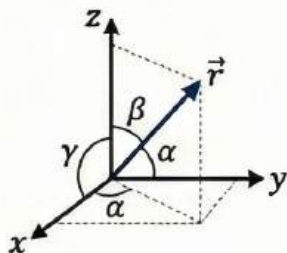


Collinear & Equal Vectors

Collinear (parallel) vectors are scalar multiples of each other ($\vec{a} = \lambda \vec{b}$), while **Equal vectors** must have the same magnitude and direction.



Direction Cosines & Ratios



Locating Vectors via Angles

Direction cosines (l, m, n) represent the cosines of angles α, β, γ that a vector makes with the x, y , and z axes.

The Fundamental Identity

$$l^2 + m^2 + n^2 = 1$$

Calculating l, m, n

$$l = \cos \alpha = x/r$$

$$m = \cos \beta = y/r$$

$$n = \cos \gamma = z/r$$

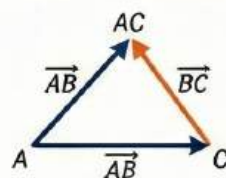
Vector Operations & Laws

Algebraic Operations

Add or subtract vectors by combining their respective $\hat{i}, \hat{j}$, and $\hat{k}$ components (e.g., $x_1 \pm x_2$).

Triangle Law of Addition

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$



Vector Joining Two Points

For the ax, 6B between points A and B:

$$\overrightarrow{AB} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Section Formula

For internal division, use $\frac{m\vec{b} + n\vec{a}}{m + n}$; for external division, change the sign to negative.

The Two Products (Dot vs. Cross)

Dot Product (Scalar)

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$$

the result is a **scalar**

Perpendicularity Condition

If two vectors are **perpendicular** ($\theta = 90^\circ$), their **Dot Product** is always zero ($\vec{a} \cdot \vec{b} = 0$).

Scalar Projection



Projection Special Cases

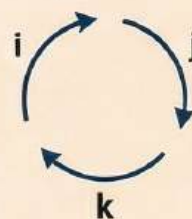
Projection is zero if vectors are perpendicular; it equals the vector itself if the angle is 0° .

Cross Product (Vector)

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n}$$

the result is a **vector** perpendicular to both $\vec{a}$ and $\vec{b}$.

$\hat{i}, \hat{j}, \hat{k}$ circle



Parallelism Condition

If two vectors are **parallel** ($\theta = 0^\circ$), their **Cross Product** is zero.

Geometric Applications & Projections

Area Calculations



$$\text{Area of Triangle} = \frac{1}{2} |\vec{a} \times \vec{b}|$$



$$\text{Area of Parallelogram} = |\vec{a} \times \vec{b}|$$

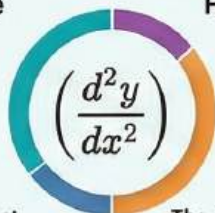
Differential Equations: Rapid Revision Guide for Class 12 Maths

A clear, step-by-step visual summary covering definitions, identification rules, and solving techniques.

FUNDAMENTALS: ORDER & DEGREE

ORDER: The Highest Derivative

Order = 2
(Highest Order Derivative)


$$\left(\frac{d^2y}{dx^2}\right)$$

The order is the highest order derivative present in the equation.

DEGREE: The Power of the Highest Order Derivative

Degree = 1
(Power of the Highest Order Derivative)

The degree is the power of the highest order derivative, provided the equation is a polynomial in derivatives.

 **When is Degree "Not Defined"?**

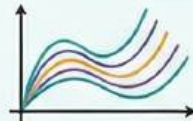
$$\log\left(\frac{dy}{dx}\right), \sin\left(\frac{dy}{dx}\right), e^{\frac{dy}{dx}}$$

Not defined if the derivative is inside a function or if powers are fractional.

TYPES OF SOLUTIONS

GENERAL SOLUTION

$$y = f(x) + C$$



Contains arbitrary constants (like +C).

The Rule of Constants

$$\text{Order} = n$$

Number of Arbitrary Constants = n

PARTICULAR SOLUTION

$$y = f(x) + 3$$



Specific values for constants based on given conditions.

Particular Solution Constant Count

0

Always contains zero arbitrary constants.

SOLVING STRATEGY: THE THREE METHODS



METHOD 1:
VARIABLE SEPARATION

$$\int f(x)dx = \int g(y)dy$$

Separate all 'x' terms with dx and all 'y' terms with dy , then integrate both sides.



METHOD 2:
HOMOGENEOUS EQUATIONS

Identified when variables appear in ratios like y/x or x/y ; solved by substituting $y = vx$.

$$f\left(\frac{y}{x}\right)$$



METHOD 3:
LINEAR DIFFERENTIAL EQUATIONS (LDE)

Identified when the dependent variable and its derivative appear only in the first power and are not multiplied together.

THE LINEAR DIFFERENTIAL EQUATION MIND MAP

$$\frac{dy}{dx} + Py = Q$$

STANDARD FORM

$$I.F. = e^{\int P dx}$$

CALCULATING THE INTEGRATING FACTOR (I.F.)

$$y \cdot (I.F.) = \int (Q \cdot I.F.) dx + C$$

FINAL SOLUTION FORMULA

TYPE 1
(Linear in y)

LINEAR DIFFERENTIAL EQUATIONS (LDE)

TYPE 2
(Linear in x)

STANDARD FORM

$$\frac{dx}{dy} + Px = Q$$

CALCULATING THE INTEGRATING FACTOR (I.F.)

$$I.F. = e^{\int P dy}$$

FINAL SOLUTION FORMULA

$$x \cdot (I.F.) = \int (Q \cdot I.F.) dy + C$$

Mastering Applications of Integrals: Rapid Revision for Class 12 Maths

A Visual Roadmap to Ace 'Area Under Curves' Problems for Board Exams

Phase 1: The Essential Toolkit

5-6 Marks Guaranteed

One major 5-mark question or case study in Board Exams



Basic Integration Proficiency

Only need basic power rules (e.g., integral of x^n) & Upper Limit - Lower Limit

The 'Golden' Formula

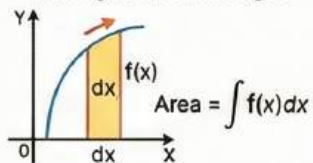
$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

Memorize to solve almost any area problem

Phase 2: Choosing Your Strip (The Method)

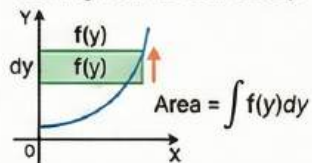
Vertical Strip (dx)

Used when the area is moving from left to right



Horizontal Strip (dy)

Used when the area is moving from bottom to top



The Modulus Rule

If area falls below X-axis, always use a modulus. Area can never be negative.

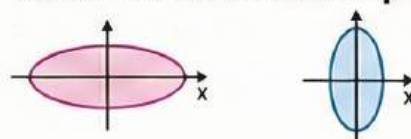
Phase 3: Identifying the Curves (Curves Types)

Circles vs. Ellipses



If coefficients of x^2 & y^2 are equal, it's a circle; if they are different, it's an ellipse

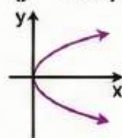
Horizontal vs. Vertical Ellipse



Larger number under x^2 = Horizontal
Larger number under y^2 = Vertical

Parabola Detection

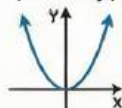
Opening Right
($y^2 = 4ax$)



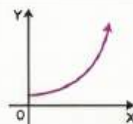
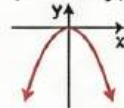
Opening Left
($y^2 = -4ax$)



Opening Up
($x^2 = 4ay$)



Opening Down
($x^2 = -4ay$)

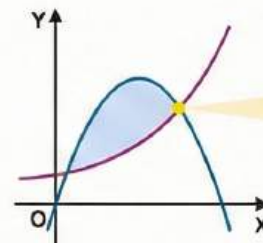


Non-Standard Curves

Use "Perfect Square" method or test $x = 0$ to find vertex for curves not passing through origin.

Phase 4: The Shading Strategy (The 'True/False' Game)

Find Points of Intersection



Always solve equations simultaneously to find meeting points before graphing

The Origin Test (0,0)

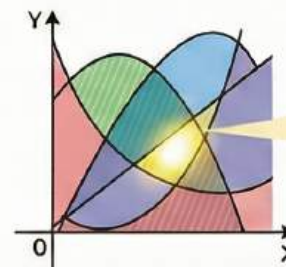


Put (0,0) into inequality

True
Shade towards origin

False
Shade away from origin

Identifying the Common Region



Final area to integrate is where shaded regions of all curves and lines overlap.

Your Ultimate Integration Cheat Sheet: A Rapid Revision Guide



The Golden Rule of Integration

Degree of 'x' must be exactly 1.
It cannot be greater, less, or fractional.

Example: $\int \cos(x^2) dx \neq \sin(x^2)$.

The Three Core Techniques

$$\int f(x) dx \rightarrow \int g'(x) dx$$

1. Integration by Substitution

$$\int f(g(x)) g'(x) dx \xrightarrow{u=g(x)} \int f(u) du$$

Simplifies a complex integral into a standard form.

2. Integration by Parts

$$\int u \cdot v dx = u \int v dx - \int (u' \cdot \int v dx) dx$$

I.L.A.T.E. Order (Choose u):

Inverse $\ln$ Logarithmic x^2 Algebraic $\sin$ Trigonometric e^x Exponential

Core Formulas (Derived from Differentiation)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \left(\frac{1}{x}\right) dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln(a)} + C$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$$

Two Powerful Result Shortcuts

$$\Rightarrow 1. \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$\Rightarrow 2. \int [f(x)]^n \cdot f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C$$

3. Partial Fractions

Degree of Numerator < Degree of Denominator

Case 1

(Distinct Linear Factors):

$$\frac{1}{(x-a)(x-b)} \rightarrow \frac{A}{x-a} + \frac{B}{x-b}$$

Case 2

(Repeated Linear Factors):

$$\frac{1}{(x-a)^2(x-b)} \rightarrow \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

Case 3

(Irreducible Quadratic):

$$\frac{1}{(x-a)(x^2+b)} \rightarrow \frac{A}{x-a} + \frac{Bx+C}{x^2+b}$$

The Problem-Solving Mind Map

Is the function **ALGEBRAIC** or **NON-ALGEBRAIC**?

ALGEBRAIC

If $N \geq D$

Always perform polynomial division first.

If $N < D$

Check if numerator is derivative of denominator ($\ln|D| + C$)

Use Partial Fractions

NON-ALGEBRAIC

1. Check the **ANGLE** (Is it linear? If not, substitute angle)
2. Is it **directly integrable**? (Use known formula)
3. If Not, can you use **substitution**? (Find function and its derivative)
4. If Still Stuck, Try **Integration by Parts** (I.L.A.T.E.) or check **Special Types**

Special Formulas & Problem Types

The "Big Three" Root Formulas

$(x/2)[\text{Question}] \pm (a^2/2)[\log \text{ or } \sin^{-1} \text{ part}]$

$$\int \sqrt{x^2 - a^2} dx \quad \int \sqrt{x^2 + a^2} dx \quad \int \sqrt{a^2 - x^2} dx$$

The e^x Shortcut

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Pattern cue:

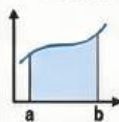
$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Special Rational Functions

$$\frac{x^2 + 1}{x^3 + kx^2 + 1} \quad \text{Divide numerator and denominator by } e^0, \text{ then use substitution}$$

Definite Integrals

Definite vs. Indefinite

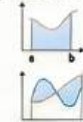


$$\int_a^b f(x) dx = F(b) - F(a) \quad (No + C)$$

Most Important Properties

Limit Swap:
 $\int_a^b f(x) dx = -\int_b^a f(x) dx$

Limit Split:
 $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$



"King's Property": $\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$
Even/Odd Property ($\int_{-a}^a f(x) dx$):
= $2 \int_0^a f(x) dx$ if even,
= 0 if odd

Mastering Applications of Derivatives: The Ultimate Class 12 Revision Guide

The Derivative as a Rate

If y is a dependent variable and x is an independent variable, $\frac{dy}{dx}$ represents how y changes with respect to x .

Positive vs. Negative Rates

+

Positive Rate:
 y increases as
 x increases

-

Negative Rate:
 y decreases as
 x decreases

Marginal Cost & Revenue

$$\text{Marginal Cost (MC)} = \frac{dC}{dx}$$

$$\text{Marginal Revenue (MR)} = \frac{dR}{dx}$$

The 4-Step Solution Map

1. Write
'Given'

2. Identify
'Find'

3. Select
'Formula'

4. Differentiate
and solve

Rate of Change of Quantities

Increasing & Decreasing Functions

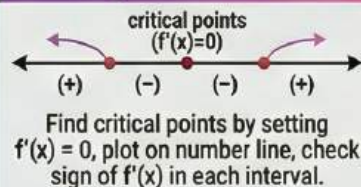
Identifying Function Behavior

Increasing:
 $f'(x) \geq 0$
Strictly Increasing:
 $f'(x) > 0$

Declining Function Values

Decreasing:
 $f'(x) \leq 0$
Strictly Decreasing:
 $f'(x) < 0$

The Sign Scheme Method



Maxima and Minima

Critical Points

Points where $f'(x) = 0$ or function is non-differentiable.

First Derivative Test (Sign Change)

Maxima

Sign changes from (+) to (-)

Minima

Sign changes from (-) to (+)

Second Derivative Test (Deeper Analysis)

Local Maxima

$f''(x) < 0$

Local Minima

$f''(x) > 0$

Absolute Maxima/Minima

For closed intervals, compare function values at all critical points and endpoints.

Point of Inflection

Sign of derivative does not change at critical point (neither maxima nor minima).

Strategy for Statement/Word Problems

Identify the Constant (Given)

Treat given info as constant to link variables

Reduce to One Variable

Before differentiating, express function in terms of single variable using Given

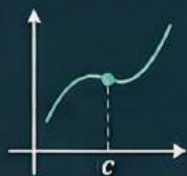
The Go-To Test

Most statement problems best solved using Second Derivative Test

Mastery of Continuity & Differentiability: The Heart of Calculus

THE FOUNDATION OF CALCULUS

Module 1: The Continuity Test



A function $f(x)$ is continuous at point c only if $LHL = RHL = f(c)$

Module 2: Algebra of Continuous Functions



If f and g are continuous, their sum, difference, product, and quotient ($g(c) \neq 0$) are also continuous.

Module 3: The Differentiability-Continuity Link



If differentiable at a point, it's guaranteed to be continuous there; reverse not always true.



Heart of Calculus

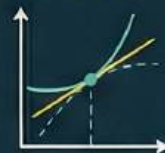
MASTER FORMULA REFERENCE

	Function $f(x)$	Derivative $f'(x)$	Note
x^n	x^n	nx^{n-1}	⏻ Power Rule
$\sim$	$\sin x$	$\cos x$	⚡ Trig Base
$\sim$	$\cos x$	$-\sin x$	"C" functions are negative
$\nless$	$\tan x$	$\sec^2 x$	⚡ Trig Base
$\nless$	$\cot x$	$-\operatorname{cosec}^2 x$	"C" functions are negative
$\nless$	$\sec x$	$\sec x \tan x$	⚡ Trig Base
$\nless$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$	"C" functions are negative
e^x	$\log x$	e^x	a^x $a^x \log a$ Constant base
	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$\ln$ Logarithmic
	$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\mathcal{P}$ Inverse Trig

"C-Rule" Mnemonic: All derivatives of C-functions (coa, cot, cosec) result in a negative sign.

DIFFERENTIABILITY ESSENTIALS

Module 1: The Differentiability Condition



A function is differentiable if $LHD = RHD$ at a specific point.

Module 2: The "Golden Rule" (Chain Rule)



1. Differentiate the Power



2. Differentiate the Function



3. Differentiate the Angle/Theta

Solve complex derivatives in these three steps sequentially.

Module 3: Logarithmic Differentiation

$y = f(x)^{g(x)}$ Used for variable-to-variable powers; take "log" on both sides to bring the power down.

THE DERIVATIVE TOOLKIT (RULES)

Module 1: Product Rule

$$u \times v$$

(First Function $\times$ Derivative of Second) + (Second Function $\times$ Derivative of First)

Module 2: Quotient Rule (u/v)

$$\frac{u}{v} = \frac{[\text{Denominator} \times \text{Deriv. of Numerator} - \text{Numerator} \times \text{Deriv. of Denominator}]}{[\text{Denominator}]^2}$$

Module 3: Parametric & Implicit Forms

Implicit:
Mixed x & y terms
e.g., $xy + x = y$

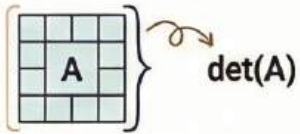
Parametric:
Uses third variable (t, θ)
e.g., $x = f(t)$
 $y = g(t)$

Determinants Rapid Revision: Master Class 12 Maths (Board 2025)

QUICK REVISION GUIDE & FORMULA SHEET

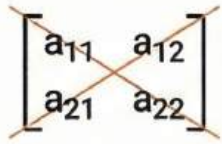
FUNDAMENTALS & CALCULATIONS

ASSOCIATED WITH SQUARE MATRICES



A determinant is a number we associate with any square matrix; if a matrix is not square, it cannot have a determinant.

CALCULATING 2X2 DETERMINANTS



$$(a_{11} * a_{22}) - (a_{12} * a_{21})$$

EXPANDING 3X3 DETERMINANTS

$$\begin{matrix} + & - & + \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{12} & a_{33} \\ a_{13} & a_{12} & a_{33} \end{bmatrix} \end{matrix}$$

Expand along any row or column (e.g., Row 1) with (+ - +) sign convention:
 $a_{11}(M_{11}) - a_{12}(M_{12}) + a_{13}(M_{13})$

MINORS, CO-FACTORS & ADJOINTS

MINORS & CO-FACTORS

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Minor (M_{ij}): Determinant after removing row/column.
Co-factor (C_{ij}): $(-1)^{i+j} * \text{Minor}$.

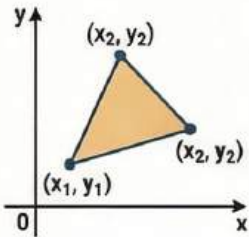
FINDING THE ADJOINT (adj A)

$$\begin{bmatrix} \text{Co-factors} & \text{Co-factors} \\ \text{Co-factors} & \text{Co-factors} \\ \text{Co-factors} & \text{Co-factors} \end{bmatrix}$$

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

Adjoint = Transpose of Co-factor Matrix.

AREA OF TRIANGLE



$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

CONDITION FOR COLLINEARITY



If calculated **Area = 0**, points are collinear (lie on the same line).

HIGH PROBABILITY 5-MARK QUESTION

THE 5-MARK MASTERY: MATRIX METHOD

SOLVING SYSTEMS ($AX = B$)

1 Calculate inverse (A^{-1})

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

Depends on Non-Singularity ($|A| \neq 0$)

2 Solve for X

$$X = A^{-1}B$$

3 Consistency Check

$|A| \neq 0 \rightarrow$ Unique Solution
 $|A| = 0$ & $(\text{adj } A)B \neq 0 \rightarrow$ Inconsistent (No Solution)

MCQ GOLDMINE: KEY PROPERTIES

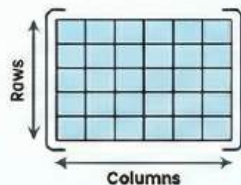
- SCALAR MULTIPLES:** $|kA| = k^n |A|$ (for order n)
- TRANPOSE EQUALITY:** $|A| = |A^T|$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & c \\ b & a \end{bmatrix}$
- SINGULAR VS. NON-SINGULAR:** Singular if $|A| = 0$; Non-Singular if $|A| \neq 0$ (Inverse exists)

QUICK REFERENCE

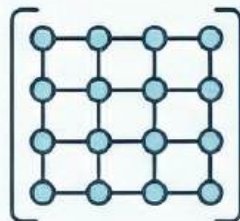
RELATIONSHIP	PROPERTY FORMULA
Product of Determinants	$ AB = A * B $
Determinant of Adjoint	$ \text{adj } A = A ^{n-1}$
Inverse Relationship	$ A^{-1} = \frac{1}{ A }$
Power Property	$ A^n = (A)^n$
Row/Co-factor Sum	$\sum (\text{Element} * \text{its own Co-factor}) = A $
Row/Co-factor Mismatch	$\sum (\text{Element} * \text{Co-factor of different row}) = 0$

Mastering Matrices: Rapid Revision for Class 12 Maths

MATRIX FUNDAMENTALS

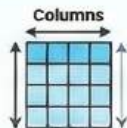


What is a Matrix?
A rectangular arrangement of numbers or functions organized into horizontal "Rows" and vertical "Columns."



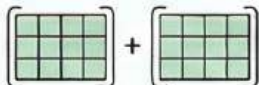
MATRIX

$R \times C$



Order of a Matrix ($R \times C$):
The dimension is always written as (Number of Rows) $\times$ (Number of Columns). Knowing the order is critical to avoid calculation errors.

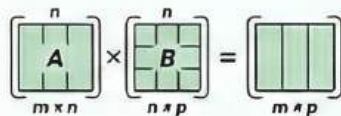
MATRIX OPERATIONS



Addition & Subtraction
Only possible if matrices have the same order: simply add or subtract corresponding elements.

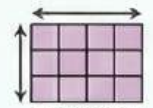


Scalar Multiplication
Multiplying a matrix by a constant (k) means multiplying every single element inside the matrix by that constant.

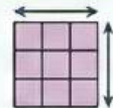


The Rule of Multiplication (AB)
To multiply A and B , the columns of A must equal the rows of B . If A is ($m \times n$) and B is ($n \times p$), the resulting matrix is ($m \times p$).

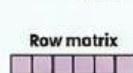
TYPES OF MATRICES



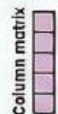
Rectangular:
Rows $\neq$ Columns



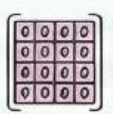
Square:
Rows = Columns
(e.g., 2×2 or 3×3)



Row matrix

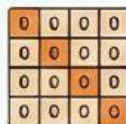


Column matrix



Zero Matrix:
A matrix where every single element is zero; it can be of any order.

THE DIAGONAL HIERARCHY



Step 1: Diagonal Matrix
A square matrix where all non-diagonal elements are zero. Diagonal elements are those where $i = j$ (e.g., a_{ii} , a_{jj}).

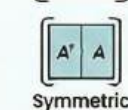


Step 2: Scalar Matrix
A diagonal matrix where all diagonal elements are the same constant value (e.g., all are 2s or 5s).

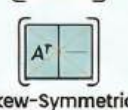


Step 3: Identity Matrix (I)
A scalar matrix where all diagonal elements are exactly 1. This is the most used matrix in calculations (denoted as I or I_n).

TRANSPOSE & SYMMETRY



Symmetric



Skew-Symmetric

Transpose of a Matrix (A^T):
Created by interchanging rows and columns. $(A^T)^T$ always returns to the original matrix A .

The Reversal Law: Crucial Exam Property: The transpose of a product reverses the order, $(AB)^T = B^T A^T$.

PRO-TIPS FOR MCQs



The Identity Power

$I^n = I$ (identity matrix raised to any power remains the identity) and $AI = IA = A$.



Skew-Symmetric Fact

All diagonal elements of a skew-symmetric matrix are always zero.



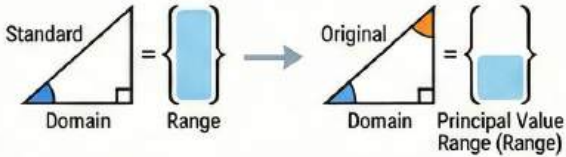
Universal Matrix

The Zero Matrix is the only matrix that is both symmetric and skew-symmetric.

Mastering Inverse Trigonometric Functions: The 15-Minute Rapid Revision Guide

Foundations & The Master Table

The Inversion Principle



Domain and Range are swapped from standard trigonometry; however, we use a restricted **"Principal Value Range"** to ensure the functions are well-defined.



Exam Weightage: 2-3 marks directly, frequently in 4-mark Case Studies, essential for Calculus.

Inverse Trig Master Table

Function	Domain	Principal Value Range (Range)
$\sin^{-1} x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	R	$(-\pi/2, \pi/2)$
$\cot^{-1} x$	R	$(0, \pi)$
$\sec^{-1} x$	$R - (-1, 1)$	$[0, \pi] - [\pi/2]$
$\operatorname{cosec}^{-1} x$	$R - (-1, 1)$	$[-\pi/2, \pi/2] - \{0\}$

The Formula Toolbox

Group 1 vs. Group 2 (Negative Angles)

Group 1

$$\sin^{-1}(-x) = -\sin^{-1} x$$

$$\tan^{-1}(-x) = -\tan^{-1} x$$

$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

Group 2

$$\cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\cot^{-1}(-x) = \pi - \cot^{-1} x - \pi$$

$$\sec^{-1}(-x) = \pi - \sec^{-1} x$$

Complementary Identities

$$(\sin^{-1} x + \cos^{-1} x) = \pi/2$$

$$(\tan^{-1} x + \cot^{-1} x) = \pi/2$$

$$(\sec^{-1} x + \operatorname{cosec}^{-1} x) = \pi/2$$

$2\tan^{-1} x$ Conversions

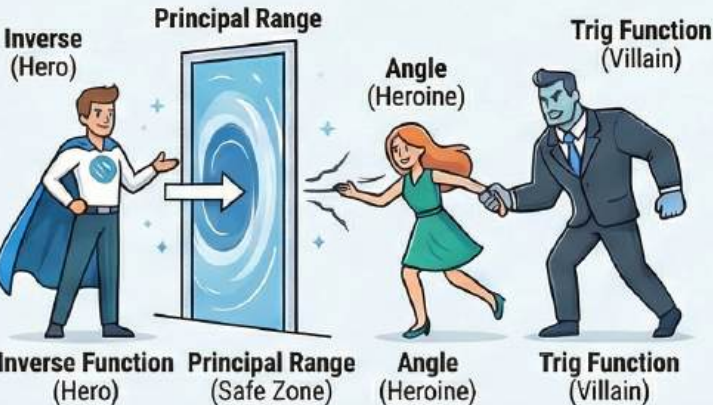
$$\sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$2\tan^{-1} x \quad \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

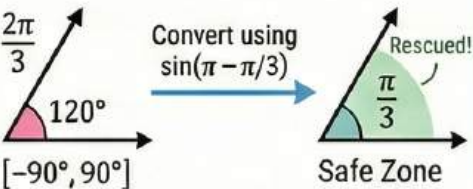
$$\tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

The "Hero-Heroine" Concept (Principal Values)

The Hero only rescues if the Heroine stays within his Principal Range conditions.



Solving $\sin^{-1}(\sin(2\pi/3))$



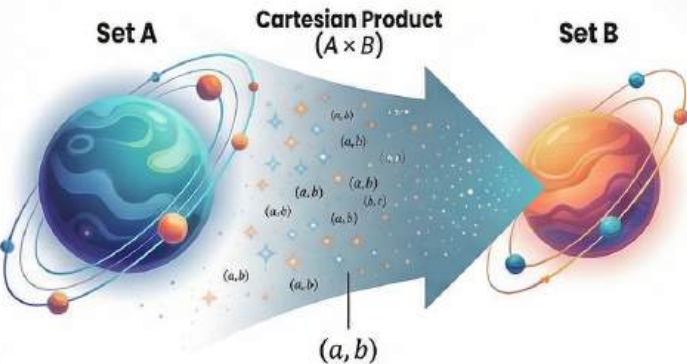
Substitution Shortcuts for Calculus

Substitution Guide

If you see...	Substitute x with...
$\sqrt{a^2 - x^2}$	$a \sin \theta$ or $a \cos \theta$
$\sqrt{a^2 + x^2}$	$a \tan \theta$ or $a \cot \theta$
$\sqrt{x^2 - a^2}$	$a \sec \theta$ or $a \operatorname{cosec} \theta$
$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$a \cos 2\theta$

Rapid Revision: Relations & Functions (Grade 12 Mathematics)

Fundamentals of Relations



A relation is a subset of the Cartesian product $A \times B$, consisting of ordered pairs (a, b) where 'a' is from the first set and 'b' is from the second.

Empty vs. Universal Relations

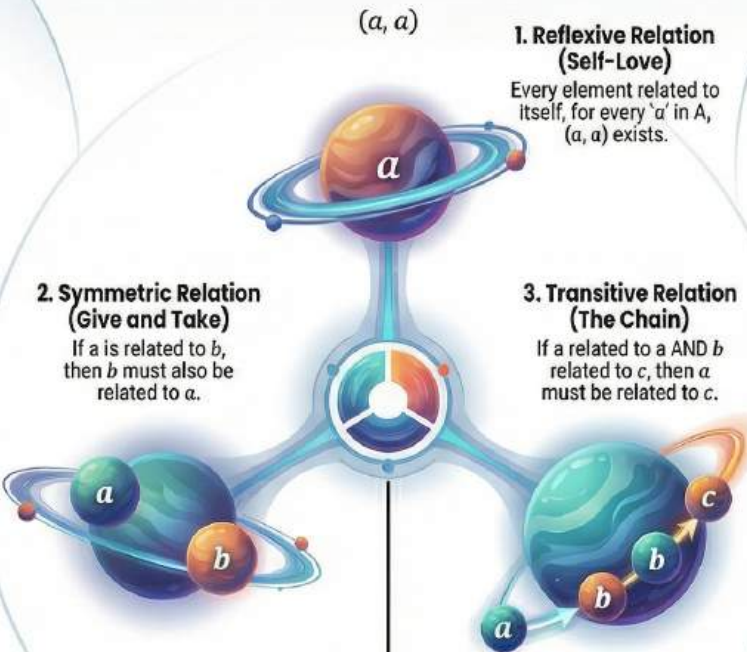


Empty (Void) Relation
No elements



Universal Relation
Contains every possible ordered pair

The Equivalence Relation Triad



1. Reflexive Relation (Self-Love)
Every element related to itself, for every 'a' in A, (a, a) exists.

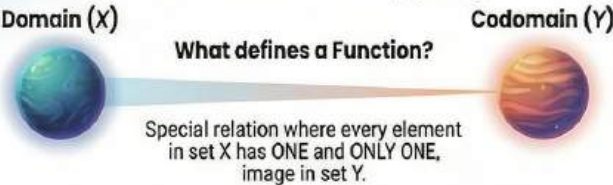
2. Symmetric Relation (Give and Take)
If a is related to b, then b must also be related to a.

3. Transitive Relation (The Chain)
If a related to a AND b related to c, then a must be related to c.

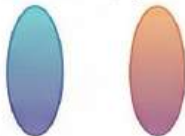
Equivalence Relation

Classified only if it satisfies ALL three conditions. Reflexive, Symmetric, and Transitive.

Functions and Mappings

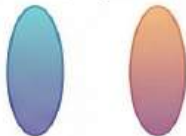


One-to-One (Injective)



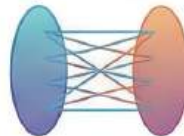
Every distinct element in domain has a distinct image in codomain.

Onto (Surjective)



Every distinct in codomain (t.l.) is image of at least one element in domain (x).

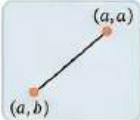
Bijective Function



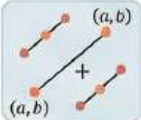
Simultaneously One-to-One and Onto.

Pro-Tips for MCQs (Beyond the Textbook)

Identity vs. Reflexive Relations



Identity Relation
ONLY allows doublets (a, b)



Reflexive Relation
Can contain additional pairs

Total Number of Functions
 n^m

Codomain size (n) raised to domain size (m).

Total Number of Functions
 $2^{(m \times n)}$

Total Number of Functions
 n^m

Codomain size (n) raised to domain size (m)

Cosmic Calculations: One-to-One & Onto

Condition	One-to-One Functions	Onto Functions
if $m < n$ (Domain smaller)	$\frac{n!}{(n-m)!}$	0
if $m > n$ (Codomain equaler)	0	(Formula based on m, n)
if $m = n$ (Equal sets)	$n!$ or $m!$	$n!$ or $m!$